INCH-POUND

MIL-C-11015/14D
14 May 2001
SUPERSEDING
MIL-C-11015/14C
6 December 1967

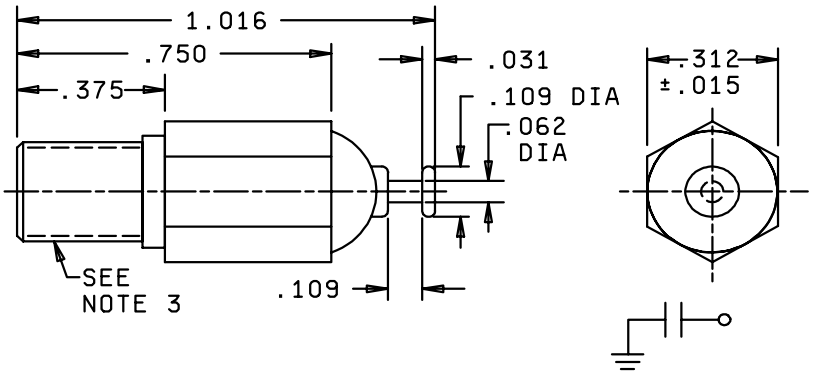
MILITARY SPECIFICATION SHEET

# CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE) (STAND-OFF), STYLE CK80

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-C-11015.

INACTIVE FOR NEW DESIGN AFTER 31 MARCH 1999
FOR REPLACEMENT PURPOSES ONLY




CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Threaded portion shall be ¼-28 UNF-2A and shall extend to within .047 (1.19 mm) of the shoulder.
4. A nut and internal tooth lockwasher shall be supplied with each capacitor.
5. Unless otherwise specified, tolerance is ±.031 (.79 mm).

| Inches | mm |
|---|---|
| .015 | .38 |
| .031 | .79 |
| .062 | 1.57 |
| .109 | 2.77 |
| .312 | 7.92 |
| .375 | 9.53 |
| .750 | 19.05 |
| 1.016 | 25.81 |

FIGURE 1. Dimensions and configuration.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REQUIREMENTS

Design and construction:

Dimensions and configuration - See figure 1.

Case type - Stand-off.

Capacitance value - See table I.

Capacitance tolerance - ±10 percent (K) or ±20 percent (M) as shown in table I.

Rated temperature - -55°C to +85°C.

Dielectric withstanding voltage: In accordance with MIL-C-11015.

Dielectric (DWV):

Test voltage - 250 percent of rated voltage, applied between the case and terminal.

Body insulation: Not applicable.

TABLE I. Style CK80 characteristics.

| PIN 1/ | Rated voltage | Rated temperature and voltage-temperature limits | Capacitance | Capacitance tolerance |
|---|---|---|---|---|
| | Volts, dc | | pF | |
| CK80AX100- | 500 | AX | 10 | K, M |
| CK80AX330- | 500 | AX | 33 | K, M |
| CK80AX470- | 500 | AX | 47 | K, M |
| CK80AX680- | 500 | AX | 68 | K, M |
| CK80AX101- | 500 | AX | 100 | K, M |
| CK80AX471- | 500 | AX | 470 | K, M |
| CK80AX681- | 500 | AX | 680 | K, M |
| CK80AW102M | 500 | AW | 1,000 | M |
| CK80AW152M | 500 | AW | 1,500 | M |

1/ Where applicable, the complete PIN will include an additional symbol to indicate capacitance tolerance.

Barometric pressure (reduced): In accordance with MIL-C-11015 and method 105 of MIL-STD-202, condition B (50,000 ft).

Test potential - 150 percent of rated voltage, applied between the case and terminal.

Insulation resistance (IR): In accordance with MIL-C-11015 and method 302 of MIL-STD-202, condition B. 20,000 megohms, minimum.

Dissipation factor (DF): 1.5 percent, maximum (voltage-temperature limit X) or 2.0 percent, maximum (voltage-temperature limit W).

Vibration, high frequency: In accordance with MIL-C-11015 and method 204 of MIL-STD-202, condition D (20 g's).

Thermal shock and immersion: In accordance with MIL-C-11015.

IR - 3,000 megohms, minimum.



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Terminal strength: In accordance with MIL-C-11015.

Moisture resistance: In accordance with MIL-C-11015.

DWV - 250 percent of rated voltage.

IR - 3,000 megohms, minimum.

Cap. - Within tolerance of table I value.

Solderability: In accordance with MIL-C-11015. 1 terminal.

Life (at elevated ambient temperature): In accordance with MIL-C-11015.

Test potential - 200 percent of rated voltage.

IR - 500 megohms, minimum (at 85°C) and 10,000 megohms, min (at 25°C).

DF - 3.0 percent, maximum (voltage-temperature limit AX) or 4.0 percent, maximum (voltage-temperature limit AW) (at 25°C).

Marking: In accordance with MIL-C-11015.

Changes from previous issue: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
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